[BUG] Incorrect shot batching behaviour in ExpvalCost when optimize=Tru

https://github.com/PennyLaneAI/pennylane/issues/1480

ROW 105

[BUG] Incorrect shot batching behaviour in ExpvalCost when optimize=True #1480

New issue

Closed   #1897

trbromley opened on Jul 27, 2021

Expected behavior

Suppose we want to evaluate the expectation value of a Hamiltonian:

```
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])
```

We want to do this using batched-shots

```
shot_batch = [(100, 10)]
```

To do this in PennyLane, we use `ExpvalCost`. There are options based on the `optimize` flag, which sets whether to group commuting observables:

```
exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)
```

Evaluating the result with `optimize=False` gives the expected output shape:

```
>>> exp_unoptimized(args)
[1.92 1.9  1.98 1.92 1.96 1.92 1.92 1.94 1.86 1.9 ]
```

Actual behavior

When `optimize=False`, we get an unexpected output shape:

```
>>> exp_optimized(args)
[1.8 2. ]
```

Additional information

We may decide not to fix this given `ExpvalCost` may soon be replaced, and suggest users set `optimize=False`. On the other hand, `optimize=True` provides some impressive speedups and switching to `optimize=False` will likely prevent users being able to successfully optimize.

Related post: https://discuss.pennylane.ai/t/differentiable-circuit-with-sampling/1190

Source code

```
import pennylane as qml

shot_batch = [(100, 10)]
dev = qml.device("default.qubit", wires=2, shots=shot_batch)
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])

def ansatz(params, wires):
    qml.RX(params, wires=wires[0])

exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)

print(exp_unoptimized(0.4))
print(exp_optimized(0.4))
```

Tracebacks

No response

Assignees
antalszava
rmoyard

Labels
bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
`ExpvalCost` when `optimize=True` with shots batch
PennyLaneAI/pennylane

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Participants

System information

```
Happens on dev branch of PL.
```

- [x] I have searched exisisting GitHub issues to make sure the issue does not already exist.

trbromley added bug on Jul 27, 2021

github-actions assigned antalszava on Jul 27, 2021

trbromley on Jul 27, 2021 Contributor Author

Do we need to edit the template to automatically format to Python? See below:

Source code

```
import pennylane as qml

shot_batch = [(100, 10)]
dev = qml.device("default.qubit", wires=2, shots=shot_batch)
h = qml.Hamiltonian([1, 1], [qml.PauliZ(0), qml.PauliZ(1)])

def ansatz(params, wires):
    qml.RX(params, wires=wires[0])

exp_unoptimized = qml.ExpvalCost(ansatz, h, dev, optimize=False)
exp_optimized = qml.ExpvalCost(ansatz, h, dev, optimize=True)

print(exp_unoptimized(0.4))
print(exp_optimized(0.4))
```

Otherwise, template was fun to use! The "System information" felt a bit unnecessary for this post, but I can understand why it's there as a required field.

mariaschuld on Jul 28, 2021 Contributor

@trbromley we are busy finding a better way to integrate `shot_batch`. Basically, it will always be ambivalent if we set it in the device, but use the device to do all sorts of things. The solution we are exploring is to do the grouping and shot distribution when creating the Hamiltonian (otherwise using a default).

So I personally suggest fixing this with the new VQE design!

🚀 2

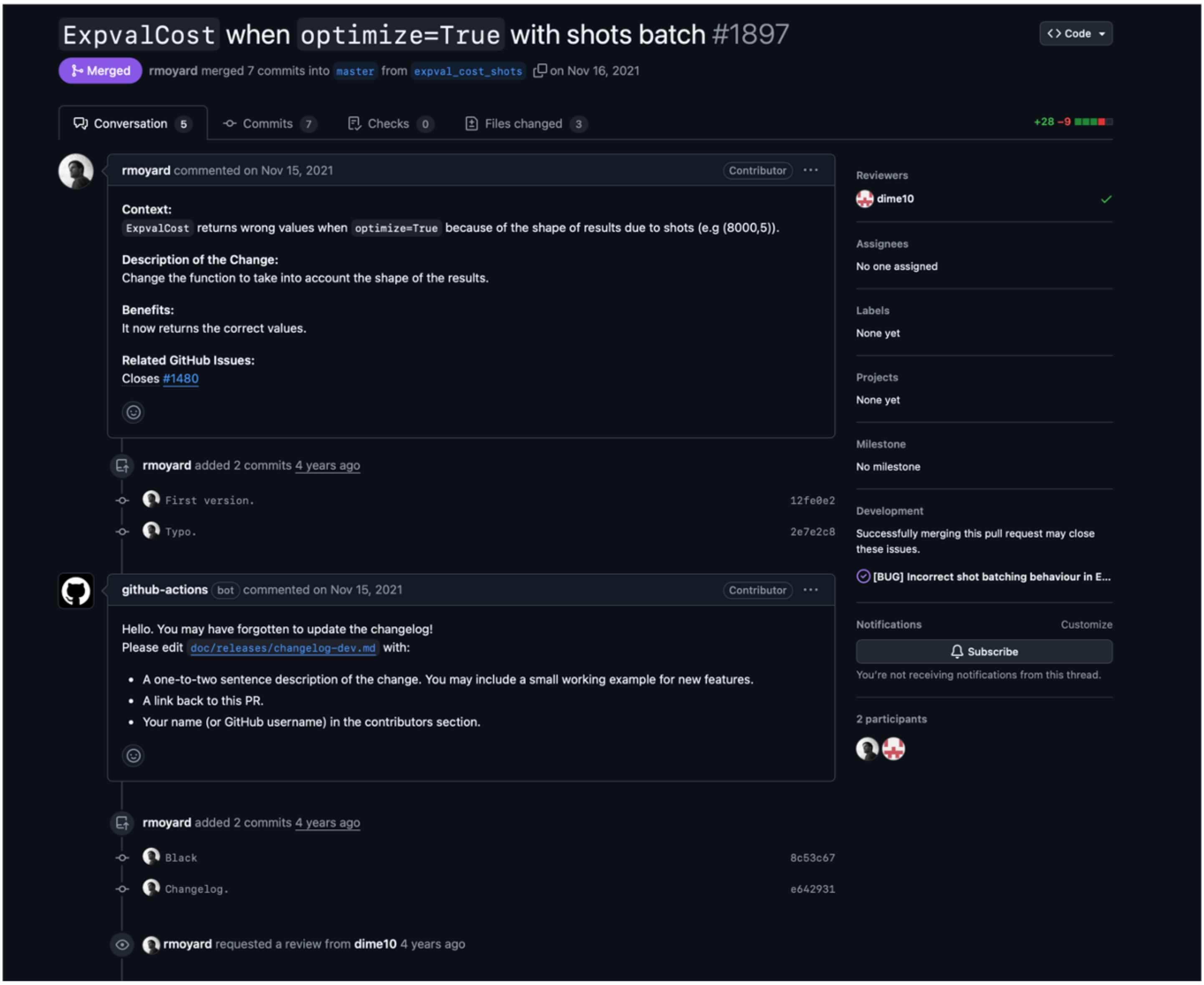
rmoyard self-assigned this on Aug 20, 2021

rmoyard mentioned this on Nov 15, 2021

`ExpvalCost` when `optimize=True` with shots batch #1897

rmoyard on Nov 15, 2021 Contributor

[sc-8002]



`ExpvalCost` when `optimize=True` with shots batch #1897

Code

Merged rmoyard merged 7 commits into `master` from `expval_cost_shots` on Nov 16, 2021

Conversation 5 | Commits 7 | Checks 0 | Files changed 3 | +28 −9

rmoyard commented on Nov 15, 2021 — Contributor

Context:
`ExpvalCost` returns wrong values when `optimize=True` because of the shape of results due to shots (e.g (8000,5)).

Description of the Change:
Change the function to take into account the shape of the results.

Benefits:
It now returns the correct values.

Related GitHub Issues:
Closes #1480

rmoyard added 2 commits 4 years ago

- `First version.` 12fe0e2
- `Typo.` 2e7e2c8

github-actions bot commented on Nov 15, 2021 — Contributor

Hello. You may have forgotten to update the changelog!
Please edit `doc/releases/changelog-dev.md` with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.

rmoyard added 2 commits 4 years ago

- `Black` 8c53c67
- `Changelog.` e642931

rmoyard requested a review from **dime10** 4 years ago

Reviewers
dime10

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[BUG] Incorrect shot batching behaviour in E...

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2 participants